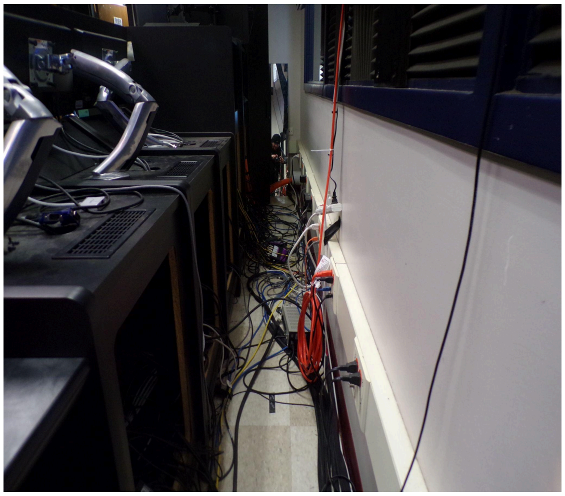
Cable Cew Portfolio

Element B

Element B: Documentation and Analysis of Prior Design Attempts

Cables from 2024-25 Engineering Design Class #1



This is the current situation with the cables. Mr O'Farrell wants us to design a cover for his cables and he wants to be able to walk on it. Our design will mainly be used by Mr O'Farrell, the Walpole Media will also be using our design but not as much as Mr. O'Farrell

Prior Design Attempt #1 - <High Capacity Cable Trough>

This is a High Capacity Cable Trough. Its dimensions are 5.25" high by 5.5" deep cable trough. This is one of the better cable troughs due to it matching our material, which is acrylic/ plexiglass. This is a very sturdy and strong material. The cable trough being transparent is very convenient in order to be able to see the wires below.

Prior Design Attempt #2 - <GRP Cable Troughing>

This is a GRP cable troughing that is 150 and 250 x 150MM. The first designs in a project are always the worst. This design will help us learn from our mistakes and improve our design to make it the best that it can be. The problem with this design is that it is too small and Mr. O'Farrell won't be able to walk on this cover due to the design.

Prior Design Attempt #3 - <Portifera Morland Protect Rubber**>**

This is the Portifera Morland Protect Rubber. It has a length of 47¼in x 15⅝in. This design is a different approach from what he had originally planned. Instead of being a cable trough it is a rubber mat that will be used to cover the cables. The rubber mat is a good approach for a different design. It will allow people to step and walk on it with it's flexibility and compression strength.

Prior Design Attempt #4 - <Cable Trench>

Had better functionality than previous attempts meaning the design provides a dedicated pathway for cables. This prevents complete disorder and minimizes exposure to external elements. The material durability is also more efficient. The reinforced metal offers life span and basic protection. This design is also very simple to access and the installation was straightforward having no excessive customization or modification. Since the design was minimal it's cost effective.

Work Cited

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